APS ELN Student Workflow Cheat Sheet

Web Book: http://10.109.92.41/MasteringELN/
PDF Book: http://10.109.92.41/MasteringELN/Mastering-ELN.pdf

Rule 1: Full tree first
Create `Procedure -> Specimen -> Action` before measuring. Never leave an action empty.

Rule 2: Raw files only
Attach native files (`.depo.json`, `.fibps`, `.mit`, `.zrw`, `.xrdml`, `.xrcx`), never screenshots.

Rule 3: Clean filenames
Name files by sample and spot (`1.fibps`, `wear 1.fibps`). Never put result numbers in filenames.

Rule 4: Extract, don't type
Use **Extract Values** (`Reader`) or **Add > Derived Value...** (`Derived`). Keep 0 `Manual` results.

WHEN & STEP	MENU / SHORTCUT (OR MCP TOOL)	WHAT TO CHECK	CH.
BEFORE THE RUN **1. Open notebook**	**File > New Notebook...** / **Open Notebook...** or Team Server login	In Community, always back up and move the `.sqlite` file and its `<name> files` folder together.	2
BEFORE THE RUN **2. Pick project & folder**	Left **Projects** panel (`Ins` New Project, `Ctrl+Shift+D` New Folder)	Put practice runs inside your personal `Practice` group; group related samples in one folder.	3, 5
BEFORE THE RUN **3. Build full sample tree**	**Item > New Procedure** → **New Specimen** → **New Action** (or **Item > New from Template...**)	Put synthesis/deposition as root `Procedure`, sample as `Specimen`, and tests as child `Actions`. Nest `Wear track` under `Friction test` and `XRR fit` under `XRR-1`.	4, 6
AT THE INSTRUMENT **4. Attach raw files**	**Item > Add File** (`Ctrl+Shift+F` in Team) or MCP `upload_file`	Attach both `.depo.json` and its paired `.tsv` log to the procedure. Verify compound file types in **Files to attach**.	4, 7
AT THE INSTRUMENT **5. Complete protocol**	**Item > Add Protocol** (`Ctrl+Shift+P`) or phone **Quick Protocol** web app	If the template uses `values.all` AutoFill, extract values first. Move status from `Draft` to `In Review` when done.	8
AFTER MEASURING **6. Extract & derive values**	**Extract Values** / right-click **Extract values from N files...** & **Add > Derived Value...**	Check step rectangles (`.fibps`), ticked indents (`.mit`), and click **Apply** on the cycle interval (`.zrw`) before extracting. Tick **Remember choices for this reader**.	4, 9
WRITE-UP & ANALYSIS **7. Link hypothesis & Wiki**	**Project > Hypotheses** (`Attach to hypothesis...`, `Add evidence...`) & **Project > New Wiki**	Cite specific files or values on the **Evidence** tab (stores MD5/value snapshot). Cross-reference records in Markdown with `[[item:N]]` and `[[specimen:CODE]]`.	10, 11
WRITE-UP & ANALYSIS **8. Search & correlate**	**Navigate > Search...** (`Ctrl+F`, tick **Search by meaning**) & **Tools > Analysis...**	**Tools > Analysis...** pairs root `Procedure` Conditions (X) with child `Action` Results (Y).	12, 13
BEFORE REVIEW **9. Audit & sign**	**Tools > Data Completeness...**, right-click **Check Value Sources...**, then **Signature > Sign Item...**	Confirm 0 missing files/values and 0 value drift *before* signing. Electronic signatures (**SIGNED**) lock the item permanently.	14, 20
WITH AN AI AGENT **10. Automate via MCP**	`ELNCommunityMCP.exe` or `ELNMCPServer.exe` (`extract_file_key_values`, `add_values`, `check_value_drift`)	Follow the **Value Route**: pass reader fields and `analysis_snapshot` verbatim to `add_values`. Save X-Ray Calc fits with `curves.job_id` so `[FITRESULT]` and `Chi2` are stored.	15–18, 21

PRE-SUBMISSION CHECK (BEFORE ASKING FOR SUPERVISOR REVIEW)

- **Tree & codes:** Sample code (e.g. `261015A` or `261014B`) matches across Procedure, Specimen, Actions, and filenames.
- **Inline summaries:** Repeated scans show `mean ± SD (n=k)` on the action row (e.g. `n=3` for 3 `.fibps` scans).
- **Completeness: Tools > Data Completeness...** (or **Data Quality Check...**) reports **0 missing** files or values.
- **Provenance & storage: Check Value Sources...** (`check_value_drift`) reports **0 drifted** values and `check_storage` reports all files **ok**.

ESSENTIAL KEYBOARD SHORTCUTS

`Ctrl+F`	Search across notebook (text & meaning)
`Ctrl+G`	Go to Item by ID or `eln://` URL
`Ctrl+Up / Down`	Reorder root procedure in Items tree
`Ctrl+Click`	Multi-select files for batch extraction
`Ctrl+Shift+F / P / V`	Add File / Protocol / Value (Team client)
`Ctrl+S / Ctrl+V`	Save Markdown / paste image attachment
`F1`	Open Keyboard Shortcuts reference

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